TRACE-Fin: Evidence-Grounded Financial Investigation with Multi-Hop Retrieval, Temporal Risk Analysis, and Symbolic Verification

A Research Framework and Evaluation Protocol

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Abstract

Financial investigations often require analysts to connect evidence dispersed across narrative reports, tables, and time-ordered transaction records. A language model that produces a fluent answer is not sufficient: investigators also need verifiable source attribution, numerically correct calculations, sensitivity to rare suspicious behavior, and a clear indication of uncertainty. This paper proposes TRACE-Fin (Trustworthy Retrieval, Analysis, and Citation Evidence for Finance), a modular framework that combines (i) query decomposition and iterative document retrieval, (ii) temporal transaction-risk scoring with feature- and time-level attention, (iii) adaptive routing to symbolic operations for numerical claims, and (iv) claim-level evidence links with calibrated abstention. The design synthesizes multi-hop financial question answering, hierarchical temporal anomaly detection, and modular symbolic-neural reasoning, while adapting them to a single auditable investigation workflow. We define the architecture, its evidence representation, learning objectives, falsifiable hypotheses, and an evaluation protocol using separate public benchmarks for transaction classification and financial document reasoning, supplemented by a newly annotated cross-source investigation set. No new experiments are reported in this manuscript; source-paper metrics are shown only as contextual evidence and are not presented as TRACE-Fin results. The proposed protocol emphasizes temporal leakage prevention, minority-class performance, answer correctness, citation support, calibration, and operational cost. TRACE-Fin is intended as decision support for human investigators, not as an autonomous determination of criminality.

Keywords: financial investigation; retrieval-augmented generation; multi-hop reasoning; anomaly detection; symbolic verification; evidence attribution; large language models

1 Introduction

Financial investigations are evidence-integration problems. An analyst may need to identify a transaction pattern, locate a related explanation in a report, combine values from a table, and explain how each source supports a conclusion. These tasks span different representations and

time scales. The growth of public blockchain data, for example, enables transaction-level analysis but does not remove challenges such as pseudonymous entities, evolving behavior, and sparse illicit labels [4]. Financial reports present a complementary challenge: a question may require locating both a textual statement and one or more table cells before performing arithmetic [6, 7].

Recent language models and retrieval-augmented generation (RAG) systems can retrieve external material and synthesize answers, but grounding alone does not ensure that each claim follows from its cited evidence or that a calculation is correct [5]. Multi-hop retrieval can connect information across sections and documents, while explicit reasoning tools can improve the handling of multi-step tasks [2, 8]. For financial use, such systems must also distinguish a model-generated risk signal from verified evidence and must expose uncertainty rather than convert incomplete information into an unqualified accusation.

This paper develops TRACE-Fin, a proposed framework for auditable financial investigation. Its design draws on three uploaded papers: Huang et al.'s GAM-CoT Transformer for temporal blockchain anomaly detection [1]; Huang et al.'s multi-hop LLaMA 3 RAG design for document-level question answering [2]; and Huang et al.'s MetaMath-LLaMA proposal for modular, interpretable multi-step reasoning with symbolic-neural components [3]. These works address distinct parts of the larger problem. TRACE-Fin makes their complementarity explicit: temporal modeling produces risk evidence; retrieval supplies documentary evidence; and a reasoning controller checks whether a proposed answer is adequately supported.

The paper makes four contributions:

1. It specifies a modular workflow that links transaction sequences, financial documents, and answer-level evidence without treating any one signal as proof of wrongdoing.
2. It defines a typed evidence ledger and a symbolic verification path for numerical and relational claims.
3. It proposes evaluation questions, baselines, metrics, and leakage controls suitable for testing each component and the integrated system.
4. It distinguishes the proposed system from empirical findings: the present manuscript reports no newly run experiments and introduces no fabricated benchmark scores.

2 Related Work and Design Foundations

2.1 Temporal transaction analysis

Blockchain AML research has studied both transaction-level classification and graph-based forensic methods. The Elliptic dataset provides time-indexed Bitcoin transaction data and has supported comparative study of conventional and graph-learning approaches [4]. The uploaded GAM-CoT paper proposes a Global Attention Module (GAM) to reweight feature channels and time steps, together with a contextual block for short-range temporal dependencies [1]. TRACE-Fin uses that decomposition as a design antecedent, but treats its output as a risk-ranking signal rather than a final investigative finding. The published metrics in that paper are not reused as results for TRACE-Fin.

2.2 Financial RAG and multi-hop retrieval

RAG combines a language model with non-parametric retrieved evidence, improving access to updateable external knowledge and supporting attribution to retrieved passages [5]. Financial QA benchmarks such as FinQA and TAT-QA test questions over financial reports, including numerical reasoning over text and tables [6, 7]. The uploaded multi-hop RAG paper describes iterative retrieval, context fusion, and generation for long-document question answering [2]. TRACE-Fin extends this pattern by tracking evidence coverage across hops and requiring every material answer claim to resolve to source spans or structured records.

2.3 Modular and symbolic reasoning

Chain-of-thought prompting established a widely studied approach for eliciting intermediate reasoning steps, but a textual reasoning trace alone does not guarantee the validity of a conclusion [8]. Tool-using approaches such as ReAct interleave reasoning and actions, while financial QA work has also used executable reasoning programs over tables and text [9, 7]. The uploaded MetaMath-LLaMA paper proposes a scheduler, a semantically grounded symbolic parser, and a hybrid symbolic-neural computation unit for interpretable mathematical reasoning [3]. TRACE-Fin adapts these ideas to select among retrieval, temporal analysis, and deterministic calculation, while storing verification outcomes separately from free-form model explanations.

Table 1: How the three uploaded papers inform TRACE-Fin. The source works motivate components; their reported performance is not transferred to this proposal.

Design source	**Relevant idea**	**TRACE-Fin adaptation**
GAM-CoT Transformer [1]	Feature- and time-aware attention for transaction sequences	Produce temporally contextualized risk scores; retain score provenance and avoid interpreting a score as a factual allegation.
Multi-hop financial RAG [2]	Dense retrieval, iterative reasoning, and context fusion	Retrieve across reports and records until evidence coverage is sufficient or the system abstains.
MetaMath-LLaMA [3]	Adaptive task routing, symbolic parsing, and hybrid reasoning	Route arithmetic and constraint checks to deterministic tools; return auditable intermediate operations and validation status.

2.4 Reported baseline comparison in the GAM-CoT source study

The uploaded GAM-CoT paper reports a comparison of six classifiers on its blockchain transaction task. Figure 1 visualizes the paper's reported precision, recall, and F1 values for the minority class. Within that table, GAM-CoT has the highest reported recall (0.932) and F1 score (0.936) among the listed models, while Random Forest has higher reported precision (0.965 versus 0.939). This is a descriptive summary of one source paper's table, not a replication or a result for TRACE-Fin. The source's evaluation details and data split are not independently verified here, so these values should be treated as preliminary context rather than evidence of generalizable superiority.

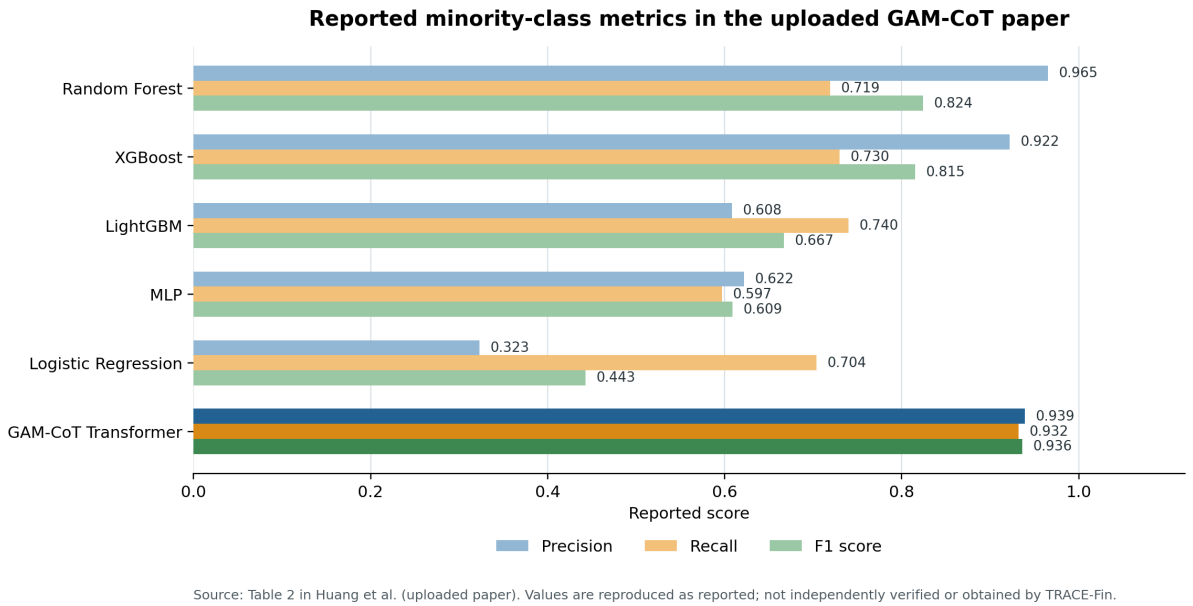


Figure 1: Minority-class metrics reproduced from Table 2 of the uploaded GAM-CoT paper [1]. Values are source-reported and are not independently verified or generated by TRACE-Fin.

3 Problem Formulation

Let an investigation request be q. The available evidence consists of a document collection $\mathcal{D} = \{d_j\}$, transaction sequences $\mathcal{X} = \{X_i\}$, and optional structured tables $\mathcal{T} = \{T_k\}$. A system response is a set of answer claims $A = \{a_m\}$ accompanied by a support ledger $\mathcal{E} = \{(a_m, e_m, v_m)\}$, where e_m identifies the supporting evidence and v_m records validation status. The system must either return claims supported by retrieved evidence and valid operations or abstain from claims for which support is insufficient.

We define the task as producing a response that jointly optimizes answer utility, source support, calculation validity, and calibrated uncertainty:

$$\max_{\hat{A}} \; U(\hat{A}, q) - \lambda_u U_{\text{unsupported}}(\hat{A}) - \lambda_c C_{\text{invalid}}(\hat{A}) - \lambda_a C_{\text{abstain}}(\hat{A}), \tag{1}$$

where U measures task usefulness, $U_{\text{unsupported}}$ penalizes unsupported claims, C_{invalid} penalizes invalid calculations or constraints, and C_{abstain} captures the cost of withholding an answer when adequate evidence exists. These terms are evaluated separately rather than collapsed into a single accuracy score.

4 TRACE-Fin Framework

4.1 Architecture overview

TRACE-Fin has five cooperating modules: an evidence normalizer, an investigation planner, a temporal risk encoder, a multi-hop retriever, and a verifier with a response composer. Evidence

identifiers persist from ingestion through the final answer. Figure 2 summarizes the flow.

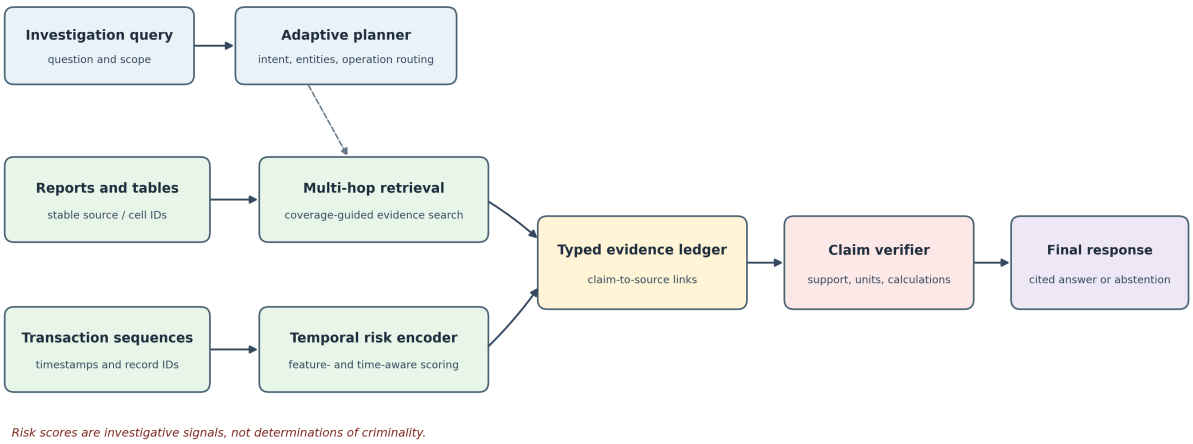


Figure 2: Proposed TRACE-Fin workflow. The planner routes questions to document retrieval and temporal risk analysis. Their outputs remain distinct evidence types until claim verification. Risk scores are investigative signals, not determinations of criminality.

4.2 Evidence normalization and planning

Each ingested item receives a stable identifier and typed metadata: source name, source location, timestamp or reporting period, entity identifiers, unit (if numeric), and extraction confidence. Text chunks retain document and page/section offsets. Table cells retain row, column, unit, and source-page references. Transaction records retain event time and the identifier of the underlying dataset record. The normalizer does not infer an entity match solely from similar names; uncertain links are marked for review.

The planner maps a query to a set of typed subquestions $\mathcal{Q} = \{q_1, \ldots, q_s\}$ and selects required operations $\mathcal{O}$. Example operation types include narrative lookup, table lookup, temporal pattern analysis, entity resolution, and arithmetic. The planner may propose a route, but cannot itself mark a claim as verified. The route is auditable and can be overridden by a human investigator.

4.3 Temporal transaction-risk encoder

For transaction entity i, let $X_i \in \mathbb{R}^{L \times F}$ denote a sequence of L observations and F features. A projection gives $H_i = X_i W_e$. To express global feature relevance and time-step relevance separately, TRACE-Fin defines gates

$$g_f = \sigma(W_f \, \mathrm{mean}_t(H_{i,t}) + b_f)\,, \tag{2}$$

$$g_t = \sigma(H_i w_t + b_t)\,, \tag{3}$$

$$\widetilde{H}_{i,t} = H_{i,t} \odot g_f \odot g_t. \tag{4}$$

A contextual block then captures local sequential patterns, for example with grouped temporal convolution and a residual connection:

$$Z_i = \mathrm{LayerNorm}\Big(\widetilde{H}_i + \mathrm{GConv}(\widetilde{H}_i)\Big), \qquad r_i = \sigma\Big(w_r^\top \,\mathrm{Pool}(Z_i) + b_r\Big). \tag{5}$$

Here r_i is a model-estimated risk score, not a probability of criminal conduct unless it has been calibrated for that interpretation. Training should use time-aware splits and class-sensitive metrics. When labels are available, a weighted binary cross-entropy can be used; the chosen threshold must be selected on validation data and reported with its operating characteristics.

4.4 Multi-hop retrieval and context fusion

The retriever ranks candidate passages using a hybrid score combining lexical and semantic relevance:

$$s(q, d) = \alpha s_{\mathrm{lex}}(q, d) + (1 - \alpha)\,\cos(\mathbf{q}, \mathbf{d}), \tag{6}$$

where $\mathbf{q}$ and $\mathbf{d}$ are query and passage embeddings, and α is selected using training/validation data. The initial retrieval set supports the planner's first subquestions. After each hop h, the system compares the current evidence ledger with the required slots; unresolved slots generate follow-up queries $q^{(h+1)}$. Retrieval stops when all required slots are supported, a hop budget is reached, or additional retrieval fails to improve coverage.

For retrieved passages d_j with relevance score s_j, the context fusion layer uses normalized weights

$$\beta_j = \frac{\exp(s_j/\tau)}{\sum_k \exp(s_k/\tau)}, \qquad c = \sum_j \beta_j\, \phi(d_j), \tag{7}$$

where τ is a temperature and ϕ encodes each passage. Fusion is used to organize evidence, not to replace the original passage-level citations. Every synthesized claim must preserve links to one or more source spans.

4.5 Symbolic verification and claim-level support

The verifier decomposes a draft response into atomic claims. Each claim is typed as factual, numerical, temporal, relational, or model-derived. Factual claims require one or more source spans. Numerical claims additionally require an executable expression tree whose leaves point to values, units, and reporting periods in the evidence ledger. For an example operation a/b, validation requires the cited numerator and denominator, compatible units, and a reproducible result within a stated rounding tolerance. Temporal and entity relations are checked against source metadata and explicit linkage rules.

The support state for claim a_m is

$$v_m \in \{\texttt{supported}, \texttt{partially-supported}, \texttt{contradicted}, \texttt{unverified}\}. \tag{8}$$

Only supported claims are presented as established findings. Partially supported or unverified claims are qualified or withheld; contradictions are surfaced rather than silently resolved by the generator. A transaction-risk score may motivate a question for further investigation, but cannot by itself support the claim that a person or entity committed an offense.

4.6 Training objective and inference control

Where the necessary annotations exist, model components can be trained with task-specific objectives. A combined objective is

$$\mathcal{L} = \lambda_r \mathcal{L}_{\text{retrieval}} + \lambda_g \mathcal{L}_{\text{generation}} + \lambda_t \mathcal{L}_{\text{transaction}} + \lambda_v \mathcal{L}_{\text{verification}}, \tag{9}$$

where the terms respectively optimize evidence ranking, answer generation, transaction risk classification, and claim/operation validation. Not every dataset supplies labels for every term; unsupported objectives must be omitted rather than populated with synthetic labels presented as ground truth. At inference, a confidence policy estimates retrieval coverage and verifier status. It returns an abstention or requests human review when required evidence is missing, conflicting, or numerically invalid.

5 Research Questions and Hypotheses

The framework is evaluated through the following falsifiable questions:

- **RQ1:** Does feature- and time-aware temporal modeling improve illicit-class recall at a fixed false-positive rate over non-sequential baselines?
- **RQ2:** Does iterative, coverage-guided retrieval improve evidence recall and citation support over one-hop retrieval?
- **RQ3:** Does symbolic checking reduce arithmetic and unit errors relative to direct language-model generation?
- **RQ4:** Does the integrated workflow improve supported-answer quality without imposing unacceptable latency or excessive abstention?

We state three hypotheses for preregistered testing: **H1**, temporal attention improves minority-class recall under a controlled false-positive rate; **H2**, multi-hop retrieval increases claim-level citation precision and evidence recall on questions requiring evidence from multiple sources; and **H3**, symbolic verification lowers numerical error rates while preserving or improving calibrated selective accuracy. Each hypothesis is rejected if the corresponding paired comparison fails to show the preregistered improvement with uncertainty intervals reported.

6 Evaluation Protocol

6.1 Datasets and tasks

The evaluation separates transaction classification from document question answering because existing public benchmarks do not, by themselves, provide aligned labels for a complete cross-source investigation. The Elliptic benchmark is proposed for temporal transaction-risk analysis; FinQA and TAT-QA are proposed for numerical and hybrid text-table question answering [4, 6, 7]. An

additional cross-source set should be assembled from lawfully usable documents and transaction records, with expert-written questions, source-span annotations, entity-link labels, executable numerical programs, and adjudicated answers. Such a dataset must be newly collected and documented; no such annotations are claimed to exist in this manuscript.

Table 2: Proposed evaluation matrix. Benchmarks are assigned only to the task types they actually support.

Evaluation set	**Task**	**Primary measures**	**Key control**
Elliptic [4]	Temporal illicit-transaction classification	AUPRC, illicit-class recall at fixed FPR, macro-F1, calibration	Respect the original time structure; avoid future-to-past leakage.
FinQA [6]	Financial numerical QA	Exact match, execution accuracy, program validity	Split by report where possible; normalize units and rounding.
TAT-QA [7]	Hybrid text-table financial QA	Answer F1, exact match, evidence-cell/span recall	Score table-cell and text-span provenance independently.
New cross-source set	Multi-hop investigation QA	Supported-claim precision, evidence recall, entity-link F1, selective accuracy	Expert annotation, adjudication, held-out entities and reporting periods.

6.2 Baselines and ablations

For transaction risk, compare logistic regression and tree-based models with a sequential neural baseline and the full temporal encoder. For document QA, compare a single-pass language model, one-hop RAG, iterative retrieval without claim verification, and TRACE-Fin. For symbolic reasoning, compare direct generation with tool-verified execution. Ablations remove in turn: (i) feature/time gates, (ii) iterative retrieval, (iii) the evidence ledger, (iv) symbolic verification, and (v) abstention. All systems should share the same source corpus, model access budget, and answer format wherever feasible.

6.3 Metrics and statistical analysis

Transaction results should report AUPRC, class-specific precision and recall, macro-F1, recall at a preregistered false-positive rate, and calibration. QA results should report exact match and answer F1 alongside execution accuracy. Grounding should be measured with evidence-span recall, claim-level citation precision (the proportion of cited claims entailed by their cited evidence), and unsupported-claim rate. Operational measures should include abstention coverage, selective accuracy, retrieval and end-to-end latency, and human review time in a controlled user study.

Use at least five fixed random seeds where training randomness applies, with paired confidence intervals (for example, bootstrap intervals over test cases). Report the complete split construction and hyperparameter-selection procedure. Use temporal and entity-disjoint splits for the newly created investigation set to reduce leakage. Human evaluators should be blinded to system identity; disagreements on support judgments should be adjudicated and inter-annotator agreement

reported. The protocol should be preregistered before test-set evaluation, including thresholds and primary outcomes.

7 Expected Contributions and Limitations

TRACE-Fin is expected to make financial QA research more measurable by treating answer correctness, source support, numerical validity, risk ranking, and abstention as distinct outcomes. If validated, its modular architecture could help investigators move from an alert to a traceable explanation while preserving the difference between model-derived risk and source-established fact. These are intended contributions and hypotheses, not empirical conclusions.

Several limitations require attention. First, transaction labels can be incomplete, delayed, or biased, and apparent model performance may not generalize across time or jurisdictions. Second, document extraction and entity resolution can introduce errors before reasoning begins. Third, citations can point to relevant text without actually entailing the associated claim; citation presence alone is therefore insufficient. Fourth, multi-hop retrieval and verification add computational cost and may increase abstentions. Fifth, a public benchmark cannot fully represent the legal, operational, and privacy constraints of a real investigation. Finally, a temporal anomaly score is not evidence of intent or criminal conduct. Deployment would require data governance, audit logging, access control, human review, and jurisdiction-specific legal review.

8 Conclusion

This paper proposed TRACE-Fin, a framework for evidence-grounded financial investigation that integrates multi-hop retrieval, temporal transaction-risk analysis, symbolic verification, and claim-level provenance. Its design synthesizes complementary ideas from the three supplied papers while grounding the evaluation plan in established financial QA and blockchain benchmarks. The framework is deliberately presented as a research proposal with a falsifiable protocol, not as a completed empirical study. The next step is to implement the components, construct and annotate a cross-source evaluation set, preregister the comparisons, and assess whether the integrated system improves supported answers without obscuring uncertainty or encouraging over-reliance on automated risk scores.

References

[1] X. Huang, C. Zhao, X. Li, C. Feng, and W. Zhang, "GAM-CoT Transformer: Hierarchical Attention Networks for Anomaly Detection in Blockchain Transactions," *Journal of Emerging Applied Artificial Intelligence*, 2025.

[2] X. Huang, Z. Lin, F. Sun, W. Zhang, K. Tong, and Y. Liu, "Enhancing Document-Level Question Answering via Multi-Hop Retrieval-Augmented Generation with LLaMA 3," TechRxiv preprint, posted June 25, 2025, doi: 10.36227/techrxiv.175086164.45839186/v1.

[3] X. Huang, Z. Wang, X. Liu, Y. Tian, and Q. Leng, "Towards Interpretable and Consistent Multi-Step Mathematical Reasoning in Large Language Models," manuscript provided by the authors, 2025.

[4] M. Weber, G. Domeniconi, J. Chen, D. K. I. Weidele, C. Bellei, T. Robinson, and C. E. Leiserson, "Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics," arXiv:1908.02591, 2019.

[5] P. Lewis *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[6] Z. Chen *et al.*, "FinQA: A Dataset of Numerical Reasoning over Financial Data," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 3697–3711, 2021, doi: 10.18653/v1/2021.emnlp-main.300.

[7] F. Zhu, W. Lei, Y. Huang, C. Wang, S. Zhang, J. Lv, F. Feng, and T.-S. Chua, "TAT-QA: A Question Answering Benchmark on a Hybrid of Tabular and Textual Content in Finance," in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*, pp. 3277–3287, 2021, doi: 10.18653/v1/2021.acl-long.254.

[8] J. Wei *et al.*, "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[9] S. Yao *et al.*, "ReAct: Synergizing Reasoning and Acting in Language Models," in *International Conference on Learning Representations*, 2023.